

A New Domination Relation for Points in the Plane*

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Abstract

Inspired by the competition number of graphs, in this paper, we introduce a new domination relation for points in the plane. The new domination relation of two points is based on the result of a binary operation $\ominus$, called the o-minus operation, on those two points. The result of the o-minus operation on two points P and Q , denoted $P \ominus Q$, is also a point called an o-minus point. We say that P dominates Q if the x -coordinate value of $P \ominus Q$ is greater than that of $Q \ominus P$. Given a set $\{P_1, P_2, \dots, P_n\}$ of points, $P_1 \ominus P_2 \ominus \dots \ominus P_n$ is also an o-minus point. The convergence point of a set of points is its resulting o-minus point with the smallest x -coordinate value. We also propose an $O(n \log n)$ -time algorithm for finding the convergence point of a set of points in the plane.

Keywords: Domination; Competition graph; Competition number; Clique cover.

1. Introduction

Inspired by the competition number of graphs [1, 2], in this paper, we define a binary operation $\ominus$ on two points in the plane as follows: Let $P = (x_1, y_1)$ and

$Q = (x_2, y_2)$ be two points in the plane. Define

$$P \ominus Q = \begin{cases} (x_1, y_1 - x_2 + y_2) & \text{if } y_1 \geq x_2, \\ (x_1 + x_2 - y_1, y_2) & \text{otherwise.} \end{cases}$$

By inspection, we can find that $P \ominus Q$ yields a point in the plane. Let $R = (r_1, r_2) = P \ominus Q$ and $T = (t_1, t_2) = Q \ominus P$. We say that R and T are the *o-minus points* of sequences $\langle P, Q \rangle$ and $\langle Q, P \rangle$, respectively. Note that different sequences may result in different o-minus points. Based on the operation $\ominus$, we say that P *dominates* Q , denoted $P \succ Q$, if $r_1 > t_1$, and, in this case, we also say that Q is *dominated* by P , denoted $Q \prec P$. For the case where $r_1 = t_1$, we say that P and Q do not dominate each other and we use $P \approx Q$ to stand for it. When $r_1 \leq t_1$ (respectively, $r_1 \geq t_1$), we also use $P \preceq Q$ (respectively, $P \succeq Q$) to denote it. Given a sequence $S = \langle P_1, P_2, \dots, P_n \rangle$ of n points in the plane, the o-minus point of S , denoted $o(S) = (s_1, s_2)$, is the resulting point obtained by $P_1 \ominus P_2 \ominus \dots \ominus P_n$. Given a set of n points in the plane, an o-minus point, say (s_1, s_2) , of some sequence of those n points is called the *convergence point* if s_1 is minimum among all o-minus points of the possible sequences of those n points. The *convergence domination problem* is to find the convergence point of points in the plane.

For example, assume that there are four points $P_1 = (9, 5)$, $P_2 = (11, 4)$, $P_3 = (3, 2)$, and $P_4 = (5, 6)$ in the plane. For the sequence $S_1 = \langle P_1, P_2, P_3, P_4 \rangle$, its o-minus point can be computed as follows:

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$$\begin{aligned}
 o(S_1) &= P_1 \ominus P_2 \ominus P_3 \ominus P_4 \\
 &= (9, 5) \ominus (11, 4) \ominus (3, 2) \ominus (5, 6) \\
 &= (15, 4) \ominus (3, 2) \ominus (5, 6) \\
 &= (15, 3) \ominus (5, 6) \\
 &= (17, 6).
 \end{aligned}$$

Now we consider another sequence $S_2 = \langle P_1, P_4, P_2, P_3 \rangle$ whose o-minus point is computed as follows:

$$\begin{aligned}
 o(S_2) &= P_1 \ominus P_4 \ominus P_2 \ominus P_3 \\
 &= (9, 5) \ominus (5, 6) \ominus (11, 4) \ominus (3, 2) \\
 &= (9, 6) \ominus (11, 4) \ominus (3, 2) \\
 &= (14, 4) \ominus (3, 2) \\
 &= (14, 3).
 \end{aligned}$$

We can find that the x -coordinate value of $o(S_2)$ is smaller than that of $o(S_1)$. It can be verified that $o(S_2)$ is the convergence point of the four points in the plane.

As an application, we introduce how to apply the convergence point problem to finding the competition number of a graph. In [1], Cohen defined the competition graph for studying ecological systems. Given an acyclic digraph $D(V, A)$, the *competition graph* of D , denoted by $C_D(V, E)$, is a simple undirected graph with the same vertex set as D , and there exists an edge xy in $E(C_D)$ provided there exist vertices $x, y, v \in V(D)$ such that $\langle x, v \rangle$ and $\langle y, v \rangle$ are arcs of $A(D)$. We also say that v is a *common prey* of x and y in D . Clearly, every undirected graph G together with sufficiently many isolated vertices can form the competition graph of some acyclic digraphs [3]. The *competition number* of an undirected graph G , denoted by $k(G)$, is the smallest number of isolated vertices added to G such that the resulting graph is the competition graph of an acyclic digraph. Imagine that there are four components P_1, P_2, P_3 , and P_4 of a graph G . Assume that we already obtain $k(P_i)$, for $1 \leq i \leq 4$, which are 9, 11, 3, and 5, respectively. Note that, for each component of G , there may be more than one vertex which can be used as a common prey for other components, denoted $c(P_i)$, for $1 \leq i \leq 4$. Here we assume that $c(P_1) = 5, c(P_2) = 4, c(P_3) = 2$, and $c(P_4) = 6$, respectively. We can also use a two-component vector (x_i, y_i) to represent the above two values for each component, i.e., $x_i = k(P_i)$ and $y_i = c(P_i)$. In our example, $(x_1, y_1) = (9, 5), (x_2, y_2) = (11, 4), (x_3, y_3) = (3, 2)$, and $(x_4, y_4) = (5, 6)$. Considering only P_1 and P_2 . Assume that the processing order is $\langle P_1, P_2 \rangle$. First, after P_1 is processed, the number of added isolated

vertices is $k(P_1) = 9$ and the number of vertices can be used as common preys for P_2 is $c(P_1) = 5$. Next, since $k(P_2) = 11$ and $c(P_1) = 5$, we only need to add another $11 - 5 = 6$ isolated vertices for processing P_2 . Accordingly, the total number of added isolated vertices after processing P_1 and P_2 is $9 + 6 = 15$ while the number of common prey vertices is 4. It can be verified that the above two resulting values can be obtained by $(9, 5) \ominus (11, 4) = (9 + 11 - 5, 4) = (15, 4)$. Note that if the processing order is P_2 before P_1 , then it will be $(11, 4) \ominus (9, 5) = (11 + 9 - 4, 5) = (16, 5)$ and the number of added isolated vertices for $\langle P_2, P_1 \rangle$ is 16. Thus, finding the competition number of a set of components is equivalent to finding the x -coordinate value of the convergence point.

The rest of this paper is organized as follows. In Section 2, we introduce some properties of the operation $\ominus$. Based on the operation $\ominus$, we define the convergence point problem in Section 3. We also propose an $O(n \log n)$ -time algorithm for solving the convergence point problem. Finally, concluding remarks are given in Section 4.

2. Some properties of the operation $\ominus$

In this section, we investigate some properties of the operation $\ominus$.

Lemma 1. *The operation $\ominus$ does not have the commutative property.*

Proof. Let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be two points in the plane, and let $(r_1, r_2) = P \ominus Q$ and $(s_1, s_2) = Q \ominus P$. By definition, $r_1 = x_1$ if $y_1 \geq x_2$; otherwise, $r_1 = x_1 + x_2 - y_1$. However, $s_1 = x_2$ if $y_2 \geq x_1$; otherwise, $s_1 = x_2 + x_1 - y_2$. It is clear that r_1 may not be equal to s_1 . This completes the proof. $\square$

Lemma 2. *The operation $\ominus$ has the associative property.*

Proof. Let $P = (x_1, y_1)$, $Q = (x_2, y_2)$, and $R = (x_3, y_3)$ be three points in the plane. All we have to prove is that $(P \ominus Q) \ominus R = P \ominus (Q \ominus R)$. We consider the following four cases.

Case 1. $y_1 \geq x_2$ and $y_2 \geq x_3$.

In this case, $P \ominus (Q \ominus R) = (x_1, y_1) \ominus (x_2, y_2 - x_3 + y_3) = (x_1, y_1 - x_2 + y_2 - x_3 + y_3)$. By $y_1 \geq x_2$, $(P \ominus Q) \ominus R = (x_1, y_1 - x_2 + y_2) \ominus (x_3, y_3)$. Since $y_1 \geq x_2$ and $y_2 \geq x_3$ imply $y_1 - x_2 + y_2 \geq x_3$, $(x_1, y_1 - x_2 + y_2) \ominus (x_3, y_3) = (x_1, y_1 - x_2 + y_2 - x_3 + y_3)$. Thus, in this case, $(P \ominus Q) \ominus R = P \ominus (Q \ominus R)$.

Case 2. $y_1 \geq x_2$ and $y_2 < x_3$.

In this case, $(P \ominus Q) \ominus R = (x_1, y_1 - x_2 + y_2) \ominus (x_3, y_3)$ and $P \ominus (Q \ominus R) = (x_1, y_1) \ominus (x_2 + x_3 - y_2, y_3)$. It is obvious that $y_1 - x_2 + y_2 \geq x_3$ is equivalent to $y_1 \geq x_2 + x_3 - y_2$. Consequently, if $y_1 - x_2 + y_2 \geq x_3$, then $(P \ominus Q) \ominus R = P \ominus (Q \ominus R) = (x_1, y_1 - x_2 + y_2 - x_3 + y_3)$; otherwise, $(P \ominus Q) \ominus R = P \ominus (Q \ominus R) = (x_1 + x_3 - y_1 + x_2 - y_2, y_3)$. Thus, this case holds.

Case 3. $y_1 < x_2$ and $y_2 \geq x_3$.

In this case, $(P \ominus Q) \ominus R = (x_1 + x_2 - y_1, y_2) \ominus (x_3, y_3) = (x_1 + x_2 - y_1, y_2 - x_3 + y_3)$ and $P \ominus (Q \ominus R) = (x_1, y_1) \ominus (x_2, y_2 - x_3 + y_3) = (x_1 + x_2 - y_1, y_2 - x_3 + y_3)$. Thus, $(P \ominus Q) \ominus R = P \ominus (Q \ominus R)$.

Case 4. $y_1 < x_2$ and $y_2 < x_3$.

In this case, $(P \ominus Q) \ominus R = (x_1 + x_2 - y_1, y_2) \ominus (x_3, y_3) = (x_1 + x_2 - y_1 + x_3 - y_2, y_3)$ and $P \ominus (Q \ominus R) = (x_1, y_1) \ominus (x_2 + x_3 - y_2, y_3)$. Since $y_1 < x_2$ and $y_2 < x_3$ imply $y_1 < x_2 + x_3 - y_2$, $(x_1, y_1) \ominus (x_2 + x_3 - y_2, y_3) = (x_1 + x_2 - y_1 + x_3 - y_2, y_3)$. This establishes the lemma. $\square$

Lemma 3. Let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be two points in the plane, and let $(r_1, r_2) = P \ominus Q$ and $(s_1, s_2) = Q \ominus P$. Then, $r_1 - r_2 = s_1 - s_2$.

Proof. By the definition of $P \ominus Q$, it can be derived that $r_1 - r_2 = (x_1 - y_1) + (x_2 - y_2)$ no matter what the relation between y_1 and x_2 is. It can be verified easily that $s_1 - s_2 = (x_1 - y_1) + (x_2 - y_2)$. This implies that $r_1 - r_2 = s_1 - s_2$. $\square$

Lemma 4. Let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be two points in the plane. If one of the following conditions holds, then $P \preceq Q$; otherwise, $P \succ Q$:

- (1) $y_1 \geq x_2$, $y_2 \geq x_1$, and $x_1 \leq x_2$,
- (2) $y_1 \geq x_2$, $y_2 \leq x_1$, and $y_2 \leq x_2$,
- (3) $y_1 \leq x_2$, $y_2 \geq x_1$, and $x_1 \leq y_1$, or
- (4) $y_1 \leq x_2$, $y_2 \leq x_1$, and $y_2 \leq y_1$.

Proof. Let $R = (r_1, r_2) = P \ominus Q$ and $S = (s_1, s_2) = Q \ominus P$. By definition,

$$r_1 = \begin{cases} x_1 & \text{if } y_1 \geq x_2, \\ x_1 + x_2 - y_1 & \text{if } y_1 < x_2. \end{cases}$$

and

$$s_1 = \begin{cases} x_2 & \text{if } y_2 \geq x_1, \\ x_2 + x_1 - y_2 & \text{if } y_2 < x_1. \end{cases}$$

Thus, after considering all possible combinations of conditions, we can find that $r_1 \leq s_1$, namely $P \preceq Q$, occurs in the following cases:

- (1) if $y_1 \geq x_2$ and $y_2 \geq x_1$, then $r_1 = x_1 \leq x_2 = s_1$,
- (2) if $y_1 \geq x_2$ and $y_2 \leq x_1$, then $r_1 = x_1 \leq x_2 + x_1 - y_2 = s_1$ which yields $y_2 \leq x_2$,
- (3) if $y_1 \leq x_2$ and $y_2 \geq x_1$, then $r_1 = x_1 + x_2 - y_1 \leq x_2 = s_1$ which yields $x_1 \leq y_1$, and
- (4) if $y_1 \leq x_2$ and $y_2 \leq x_1$, then $r_1 = x_1 + x_2 - y_1 \leq x_2 + x_1 - y_2 = s_1$ which yields $y_2 \leq y_1$.

This completes the proof. $\square$

Corollary 5. Let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be two points in the plane. If one of the following conditions holds, then $P \prec Q$:

- (1) $y_1 \geq x_2$, $y_2 \geq x_1$, and $x_1 < x_2$,
- (2) $y_1 \geq x_2$, $y_2 \leq x_1$, and $y_2 < x_2$,
- (3) $y_1 \leq x_2$, $y_2 \geq x_1$, and $x_1 < y_1$, or
- (4) $y_1 \leq x_2$, $y_2 \leq x_1$, and $y_2 < y_1$.

As an illustration, Figure 1 depicts the four conditions of $P \preceq Q$ in Lemma 4 in which an arc from x to y represents $x \geq y$.

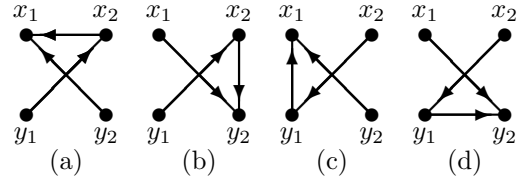


Figure 1: Illustrations for the cases in Lemma 4.

Lemma 6. Let $P = (x_1, y_1)$ be a point in the plane with $y_1 > x_1$. Then, $P \succ Q$, $P \approx Q$, and $P \prec Q$ if and only if Q is in $\{(x, y) | \min\{x_1, y\} > x\}$, $\{(x, y) | \min\{x_1, y\} = x\}$, and $\{(x, y) | \min\{x_1, y\} < x\}$, respectively.

Proof. By Conditions (1)-(4) in Lemma 4, every point Q in $\{(x, y) | x_1 < x \leq y_1, x_1 \leq y\}$, $\{(x, y) | y < x \leq y_1, y \leq x_1\}$, $\{(x, y) | x_1 < y_1 \leq x, x_1 \leq y\}$, and $\{(x, y) | y < y_1 \leq x, y \leq x_1\}$, respectively, dominates P . Since $y_1 > x_1$, the union of the first and the third sets above is $\{(x, y) | x_1 < x, x_1 \leq y\}$ while the union of the second and the fourth sets is $\{(x, y) | y < x, y \leq x_1\}$. Moreover, the union of sets $\{(x, y) | x_1 < x, x_1 \leq y\}$ and $\{(x, y) | y < x, y \leq x_1\}$ is the set $\{(x, y) | \min\{x_1, y\} < x\}$ (see Figure 2(a)). Thus, $P \prec Q$ for every point $Q \in \{(x, y) | \min\{x_1, y\} < x\}$. Similarly, we can prove that $P \approx Q$ and $P \succ Q$ for every point $Q \in \{(x, y) | \min\{x_1, y\} = x\}$ and

$Q \in \{(x, y) | \min\{x_1, y\} > x\}$, respectively. It is clear that the three sets $\{(x, y) | \min\{x_1, y\} < x\}$, $\{(x, y) | \min\{x_1, y\} = x\}$, and $\{(x, y) | \min\{x_1, y\} > x\}$ partitions the points in the plane. Therefore, the lemma follows. $\square$

Lemma 7. *Let $P = (x_1, y_1)$ be a point in the plane with $y_1 < x_1$. Then, $P \succ Q$, $P \approx Q$, and $P \prec Q$ if and only if Q is in $\{(x, y) | y > \min\{x, y_1\}\}$, $\{(x, y) | y = \min\{x, y_1\}\}$, and $\{(x, y) | y < \min\{x, y_1\}\}$, respectively.*

Proof. Since $y_1 < x_1$, we only need to consider Conditions (2) and (4) in Lemma 4. By Conditions (2) and (4) in Lemma 4, every point Q in $\{(x, y) | y < x \leq y_1\}$ and $\{(x, y) | y < y_1 \leq x\}$ dominates P . The union of sets $\{(x, y) | y < x \leq y_1\}$ and $\{(x, y) | y < y_1 \leq x\}$ is the set $\{(x, y) | y < \min\{x, y_1\}\}$ (see Figure 2(b)). Thus, $P \prec Q$ for every point $Q \in \{(x, y) | y < \min\{x, y_1\}\}$. Similarly, we can prove that $P \approx Q$ and $P \succ Q$ for every point $Q \in \{(x, y) | y = \min\{x, y_1\}\}$ and $Q \in \{(x, y) | y > \min\{x, y_1\}\}$, respectively. It is clear that the three sets $\{(x, y) | y < \min\{x, y_1\}\}$, $\{(x, y) | y = \min\{x, y_1\}\}$, and $\{(x, y) | y > \min\{x, y_1\}\}$ partitions the points in the plane. Therefore, the lemma follows. $\square$

Lemma 8. *Let $P = (x_1, y_1)$ be a point in the plane with $y_1 = x_1$. Then, $P \succ Q$, $P \approx Q$, and $P \prec Q$ if and only if Q is in $\{(x, y) | \min\{x_1, y\} > x\}$, $\{(x, y) | x_1 \leq \min\{x, y\} \text{ or } x = y\}$, and $\{(x, y) | y < \min\{x, x_1\}\}$, respectively.*

Proof. Since $y_1 = x_1$, it is easy to verify that $Q \approx P$ if and only if $Q \in \{(x, y) | x_1 \leq \min\{x, y\} \text{ or } x = y\}$ (see Figure 2(c)). By Lemmas 6 and 7, $P \succ Q$ and $P \prec Q$ if and only if Q is in $\{(x, y) | \min\{x_1, y\} > x\}$ and $\{(x, y) | y < \min\{x, x_1\}\}$, respectively. This completes the proof of this lemma. $\square$

3. An algorithm for finding the convergence point

In this section, we propose an $O(n \log n)$ -time algorithm for finding the convergence point of a set of n points in the plane.

Lemma 9. *The relation $\approx$ does not have the transitive property.*

Proof. Let $P = (x_1, y_1)$, $Q = (x_2, y_2)$, and $R = (x_3, y_3)$ be three points in the plane. We can find that $P \approx Q$ and $Q \approx R$ do not imply $P \approx R$ when the condition holds: $x_2 = y_2$, $y_1 > x_2$, $x_1 > y_2$, $y_3 > x_2$, and $x_3 > y_2$. By $x_2 = y_2$, $y_1 > x_2$, and $x_1 > y_2$, this yields $P \approx Q$. Similarly, by $x_2 = y_2$, $y_3 > x_2$, and $x_3 > y_2$, this yields $Q \approx R$. However, any relation in $\{\prec, \succ, \approx\}$ is possible for P and R . This concludes the proof of this lemma. $\square$

Lemma 10. *Both relations $\prec$ and $\succ$ have the transitive property.*

Proof. We only prove that the relation $\prec$ has the transitive property. The proof that $\succ$ has the transitive property can be handled similarly. Let $P = (x_1, y_1)$, $Q = (x_2, y_2)$, and $R = (x_3, y_3)$ be three points in the plane. All we have to prove is that $P \prec Q$ and $Q \prec R$ imply $P \prec R$.

By Corollary 5, we have the following four cases to consider.

Case 1. $y_1 \geq x_2$, $y_2 \geq x_1$, and $x_1 < x_2$.

By Corollary 5 again, we have the following four subcases to consider when $Q \prec R$.

Case 1.1. $y_2 \geq x_3$, $y_3 \geq x_2$, and $x_2 < x_3$.

In this subcase, we can find that all of x_3 , y_3 , and y_1 are greater than x_1 (see Figure 3(a)). Thus, if $y_1 \geq x_3$ (respectively, $y_1 < x_3$), then by Condition

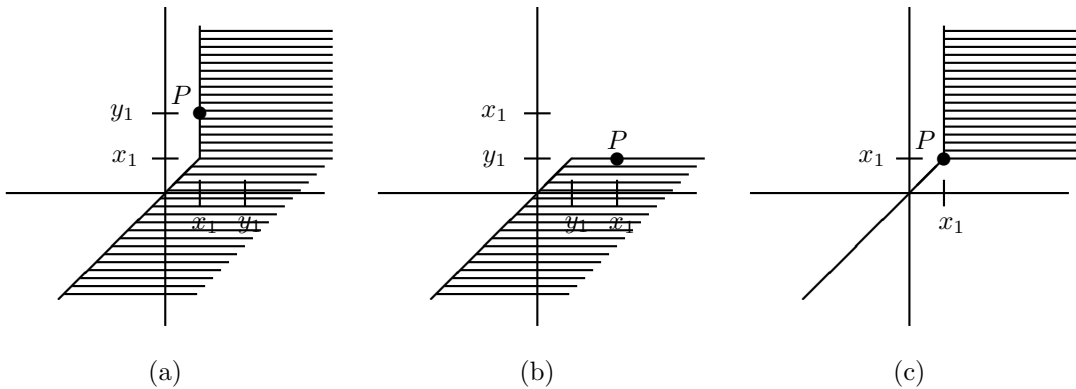


Figure 2: Illustrations for Lemmas 6-8.

(1) (respectively, Condition (3)) in Corollary 5, $P \prec R$.

Case 1.2. $y_2 \geq x_3$, $x_2 \geq y_3$, and $y_3 < x_3$.

In this subcase, we can find that $y_1 \geq x_2 > x_1$, $y_1 \geq x_2 \geq y_3$, and $x_3 > y_3$ (see Figure 3(b)). It can be verified easily that

- (1) if $y_1 \geq x_3$ and $y_3 \geq x_1$, then $x_3 > y_3 \geq x_1$ implies that Condition (1) in Corollary 5 holds,
- (2) if $y_1 \geq x_3$ and $y_3 \leq x_1$, then $x_3 > y_3$ implies that Condition (2) in Corollary 5 holds,
- (3) if $y_1 \leq x_3$ and $y_3 \geq x_1$, then $x_1 < y_1$ implies that Condition (3) in Corollary 5 holds, and
- (4) if $y_1 \leq x_3$ and $y_3 \leq x_1$, then $y_1 > x_1 \geq y_3$ implies that Condition (4) in Corollary 5 holds.

Thus, $P \prec R$ in this subcase.

Case 1.3. $y_2 \leq x_3$, $y_3 \geq x_2$, and $x_2 < y_2$.

In this subcase, we can find that all of y_1 , x_3 , and y_3 are greater than x_1 (see Figure 3(c)). Thus, by using a similar argument as in Case 1.1, $P \prec R$ holds.

Case 1.4. $y_2 \leq x_3$, $y_3 \leq x_2$, and $y_3 < y_2$.

In this subcase, we can find that $y_1 \geq x_2 > x_1$, $x_3 \geq y_2 \geq x_1$, and $x_3 \geq y_2 > y_3$ (see Figure 3(d)). It can be verified easily that

- (1) if $y_1 \geq x_3$ and $y_3 \geq x_1$, then $x_3 > y_3 \geq x_1$ implies that $x_3 > x_1$ must hold which satisfies Condition (1) in Corollary 5,
- (2) if $y_1 \geq x_3$ and $y_3 \leq x_1$, then $x_3 \geq y_2 > y_3$ implies that Condition (2) in Corollary 5 holds,
- (3) if $y_1 \leq x_3$ and $y_3 \geq x_1$, then $x_1 < y_1$ implies that Condition (3) in Corollary 5 holds, and
- (4) if $y_1 \leq x_3$ and $y_3 \leq x_1$, then $y_1 > x_1 \geq y_3$ implies that Condition (4) in Corollary 5 holds.

This establishes that $P \prec R$ in Case 1.

Case 2. $y_1 \geq x_2$, $y_2 \leq x_1$, and $y_2 < x_2$.

The cases that $y_2 \geq x_3$, $y_3 \geq x_2$, and $x_3 > x_2$ (see Figure 3(e)), and that $y_2 \leq x_3$, $y_3 \geq x_2$, and $x_2 < y_2$ (see Figure 3(g)) result in $x_2 = y_2$ which are impossible. Thus, we only need to consider the following two cases.

Case 2.1. $y_2 \geq x_3$, $x_2 \geq y_3$, and $x_3 > y_3$.

We can find that $y_1 \geq x_2 > y_2 \geq x_3$, $x_1 \geq y_2 \geq x_3 > y_3$, and $x_3 > y_3$ (see Figure 3(f)). By Condition (2) of Corollary 5, $P \prec R$.

Case 2.2. $y_2 \leq x_3$, $y_3 \leq x_2$, and $y_2 > y_3$.

We can find that all of x_1 , y_1 , and x_3 are greater than or equal to y_3 (see Figure 3(h)). Thus, if $y_1 \geq x_3$ (respectively, $y_1 < x_3$), then by Condition (2) (respectively, Condition (4)) in Corollary 5, $P \prec R$.

Case 3. $y_1 \leq x_2$, $y_2 \geq x_1$, and $x_1 < y_1$.

We have the following four subcases to consider.

Case 3.1. $y_2 \geq x_3$, $y_3 \geq x_2$, and $x_3 > x_2$.

We can find that $y_1 \geq x_1$, $x_3 > x_2 \geq y_1$, and $y_3 \geq x_2 \geq y_1 > x_1$ (see Figure 3(i)) which satisfies Condition (3) in Corollary 5. Thus, $P \prec R$.

Case 3.2. $y_2 \geq x_3$, $y_3 \leq x_2$, and $x_3 > y_3$.

In this subcase, we can only obtain that $y_1 \geq x_1$ and $x_3 \geq y_3$ (see Figure 3(j)). However, after checking all of the following combinations:

- (1) if $y_1 \geq x_3$ and $y_3 \geq x_1$, then $x_3 > y_3 \geq x_1$ implies that Condition (1) in Corollary 5 holds,
- (2) if $y_1 \geq x_3$ and $y_3 \leq x_1$, then $x_3 > y_3$ implies that Condition (2) in Corollary 5 holds,
- (3) if $y_1 \leq x_3$ and $y_3 \geq x_1$, then $x_1 < y_1$ implies that Condition (3) in Corollary 5 holds, and
- (4) if $y_1 \leq x_3$ and $y_3 \leq x_1$, then $y_1 > x_1 \geq y_3$ implies that Condition (4) in Corollary 5 holds.

Thus, $P \prec R$.

Case 3.3. $y_2 \leq x_3$, $y_3 \geq x_2$, and $x_2 < y_2$.

This subcase results in $x_1 < y_1$, $x_3 \geq y_2 > x_2 \geq y_1$, and $y_3 \geq x_2 \geq y_1 > x_1$ (see Figure 3(k)) which is Condition (3) in Corollary 5. Thus, $P \prec R$.

Case 3.4. $y_2 \leq x_3$, $y_3 \leq x_2$, and $y_3 < y_2$.

In this subcase, we can also find that $y_1 > x_1$, $x_3 \geq y_2 \geq x_1$, and $x_3 \geq y_2 > y_3$ (see Figure 3(l)). Then, by using a similar argument as in Case 1.4, we can find that $P \prec R$.

Case 4. $y_1 \leq x_2$, $y_2 \leq x_1$, and $y_2 < y_1$.

The subcases that $y_2 \geq x_3$, $y_3 \geq x_2$, and $x_3 > x_2$ and that $y_2 \leq x_3$, $y_3 \geq x_2$, and $x_2 < y_2$ result in $x_2 = y_2$ which are impossible (see Figures 3(m) and 3(o)). Thus, we only need to consider the following two cases.

Case 4.1. $y_2 \geq x_3$, $y_3 \leq x_2$, and $x_3 > y_3$.

This subcase results in $x_1 \geq y_2 \geq x_3 > y_3$, $y_1 > y_2 \geq x_3$, and $x_3 > y_3$ (see Figure 3(n)) which satisfies Condition (2) in Corollary 5. Thus, $P \prec R$.

Case 4.2. $y_2 \leq x_3$, $y_3 \leq x_2$, and $y_2 > y_3$.

We can find that all of x_1 , y_1 , and x_3 are greater than or equal to y_3 (see Figure 3(p)). Then, by using a similar argument as in Case 2.2, this subcase holds. This concludes the proof of this lemma. $\square$

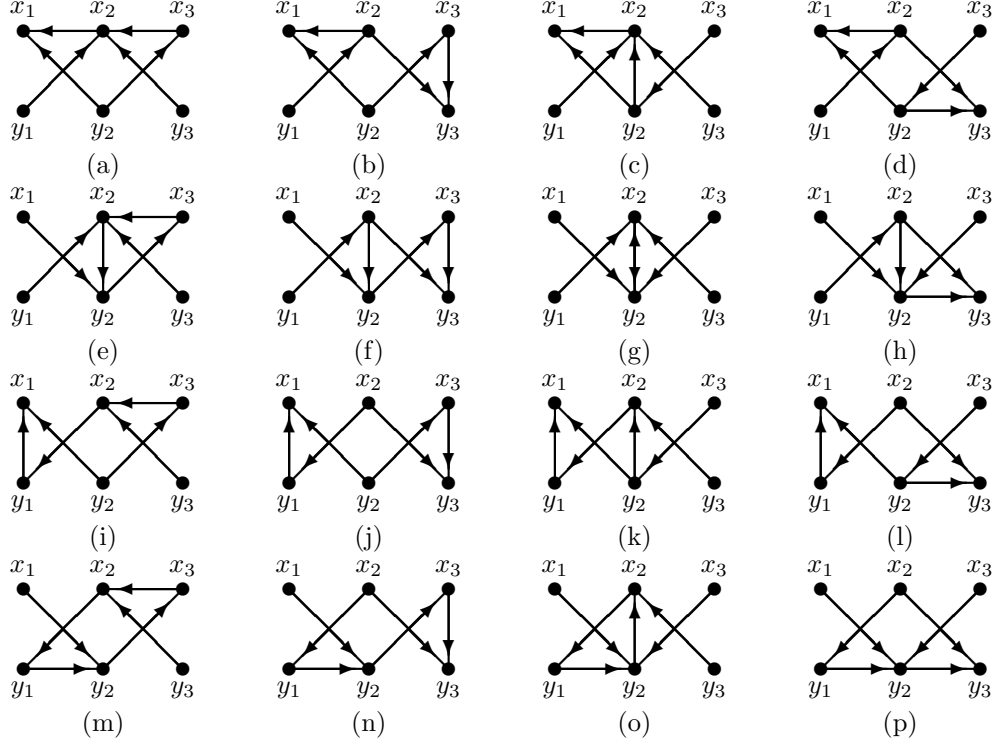


Figure 3: Illustrations for the cases in Lemma 10.

Corollary 11. Let $P = (x_1, y_1)$, $Q = (x_2, y_2)$, and $R = (x_3, y_3)$ be three points in the plane with $P \approx Q$ and $Q \approx R$. If either $x_2 \neq y_2$ or $x_2 > \max\{y_1, y_3\}$ when $x_2 = y_2$, then both relations $\lesssim$ and $\gtrsim$ have the transitive property.

Proof. If $x_2 \neq y_2$, then, by using a similar argument as in Lemma 10, this case holds. For the case where $x_2 = y_2$ and $x_2 > \max\{y_1, y_3\}$, it is easy to check that only Subcases 3.2, 3.4, 4.1, and 4.2 (see Figures 3(j), 3(l), 3(n), 3(p), respectively) satisfy the requirement of this corollary. Thus, this corollary follows. $\square$

Lemma 12. Let $P = (x_1, y_1)$, $Q = (x_2, y_2)$, and $R = (x_3, y_3)$ be three points in the plane with $P \lesssim Q$ and $Q \lesssim R$. If either $x_2 \neq y_2$ or $x_2 > \max\{y_1, y_3\}$ when $x_2 = y_2$, then $o(\langle P, Q \rangle) \lesssim R$ and $P \lesssim o(\langle Q, R \rangle)$.

Proof. We only prove that $o(\langle P, Q \rangle) \lesssim R$. With a similar reasoning, we can also prove that $P \lesssim o(\langle Q, R \rangle)$. Analogous to Lemma 10, we also have four cases to consider and, in each case, we have four subcases to consider. In Cases 1 and 2, $o(\langle P, Q \rangle) = (x_1, y_1 - x_2 + y_2)$ while, in Cases 3 and 4, $o(\langle P, Q \rangle) = (x_1 + x_2 - y_1, y_2)$. Note that, in Cases 1 and 2, $y_1 - x_2 + y_2 \geq y_2$ while, in Cases 3 and 4, $x_1 \leq x_1 + x_2 - y_1$. Now we investigate each subcase as follows.

Case 1. $y_1 \geq x_2$, $y_2 \geq x_1$, and $x_1 \leq x_2$.

In this case, $y_1 - x_2 + y_2 \geq x_1$. We consider the following four subcases.

Case 1.1. $y_2 \geq x_3$, $y_3 \geq x_2$, and $x_2 \leq x_3$.

In this subcase, we can find that $x_3 \geq x_1$, $y_3 \geq x_1$, and $y_1 - x_2 + y_2 \geq y_2 \geq x_3$ (see Figure 4(a)). Thus, by Condition (1) in Lemma 4, $o(\langle P, Q \rangle) \lesssim R$.

Case 1.2. $y_2 \geq x_3$, $x_2 \geq y_3$, and $y_3 \leq x_3$.

In this subcase, we can find that $x_3 \geq y_3$. Furthermore, since y_2 is greater than or equal to x_1 , x_3 , and y_3 , $y_1 - x_2 + y_2$ is also greater than or equal to x_1 , x_3 , and y_3 (see Figure 4(b)). Thus, if $y_3 \geq x_1$ (respectively, $y_3 < x_1$), then $x_3 \geq y_3 \geq x_1$ and by Condition (1) (respectively, (2)) in Lemma 4, $o(\langle P, Q \rangle) \lesssim R$.

Case 1.3. $y_2 \leq x_3$, $y_3 \geq x_2$, and $x_2 \leq y_2$.

In this subcase, we can find that all of y_1 , x_3 , and y_3 are greater than or equal to x_1 (see Figure 4(c)). Thus, if $y_1 - x_2 + y_2 \geq x_3$ (respectively, $y_1 - x_2 + y_2 < x_3$), then by Condition (1) (respectively, (3)) in Lemma 4, $o(\langle P, Q \rangle) \lesssim R$.

Case 1.4. $y_2 \leq x_3$, $y_3 \leq x_2$, and $y_3 \leq y_2$.

In this subcase, we can find that both $y_1 - x_2 + y_2$ and x_3 are greater than or equal to x_1 and y_3 (see Figure 4(d)). Thus, $o(\langle P, Q \rangle) \lesssim R$ no matter what the relations between x_1 and y_3 , and $y_1 - x_2 + y_2$ and x_3 are.

Case 2. $y_1 \geq x_2$, $y_2 \leq x_1$, and $y_2 \leq x_2$.

As discussed in Lemma 10, the case that $y_2 \geq x_3$, $y_3 \geq x_2$, and $x_3 \geq x_2$, and that $y_2 \leq x_3$, $y_3 \geq x_2$, and $x_2 \leq y_2$ result in $x_2 = y_2$ which contradicts the assumption that $x_2 > \max\{y_1, y_3\}$ when $x_2 = y_2$. Thus, we only need to consider the following two cases.

Case 2.1. $y_2 \geq x_3$, $x_2 \geq y_3$, and $x_3 \geq y_3$.

We can find that $y_1 - x_2 + y_2 \geq y_2 \geq x_3$, $x_1 \geq y_2 \geq x_3 \geq y_3$, and $x_3 \geq y_3$ (see Figure 4(e)). By Condition (2) of Lemma 4, $o(\langle P, Q \rangle) \lesssim R$.

Case 2.2. $y_2 \leq x_3$, $y_3 \leq x_2$, and $y_2 \geq y_3$.

We can find that x_1 and x_3 are greater than or equal to y_3 , and $y_1 - x_2 + y_2 \geq y_2 \geq y_3$ (see Figure 4(f)). Thus, if $y_1 - x_2 + y_2 \geq x_3$ (respectively, $y_1 - x_2 + y_2 < x_3$), then by Condition (2) (respectively, Condition (4)) in Lemma 4, $o(\langle P, Q \rangle) \lesssim R$.

Case 3. $y_1 \leq x_2$, $y_2 \geq x_1$, and $x_1 \leq y_1$.

We have the following four subcases to consider.

Case 3.1. $y_2 \geq x_3$, $y_3 \geq x_2$, and $x_3 \geq x_2$.

We can find that $y_2 \geq x_3$, and x_3 and y_3 are greater than or equal to $x_1 + x_2 - y_1$ since $x_2 \geq x_1 + x_2 - y_1$ (see Figure 4(g)). This implies that Condition (1) in Lemma 4 is satisfied. Thus, $o(\langle P, Q \rangle) \lesssim R$.

Case 3.2. $y_2 \geq x_3$, $y_3 \leq x_2$, and $x_3 \geq y_3$.

In this subcase, we can obtain that $y_2 \geq x_3$, $y_2 \geq y_3$, and $x_3 \geq y_3$ (see Figure 4(h)). Thus, if $x_1 + x_2 - y_1 \geq y_3$, then, by Condition (2) in Lemma 4, $o(\langle P, Q \rangle) \prec R$. For the case where $x_1 + x_2 - y_1 \leq y_3$, $x_3 \geq x_1 + x_2 - y_1$ for $x_3 \geq y_3$. Thus, by Condition

(1) in Lemma 4, $o(\langle P, Q \rangle) \lesssim R$.

Case 3.3. $y_2 \leq x_3$, $y_3 \geq x_2$, and $x_2 \leq y_2$.

In this subcase, $x_3 \geq y_2$, $y_2 \geq x_1 + x_2 - y_1$, and $y_3 \geq x_1 + x_2 - y_1$ since $x_2 \geq x_1 + x_2 - y_1$ (see Figure 4(i)). Thus, by Condition (3) in Lemma 4, $o(\langle P, Q \rangle) \lesssim R$.

Case 3.4. $y_2 \leq x_3$, $y_3 \leq x_2$, and $y_2 \geq y_3$.

In this subcase, we can also find that $x_3 \geq y_2$, $x_3 \geq y_3$, and $y_2 \geq y_3$. If $x_1 + x_2 - y_1 \geq y_3$ (see Figure 4(j)), then, by Condition (4) in Lemma 4, $o(\langle P, Q \rangle) \lesssim R$. For the case where $x_1 + x_2 - y_1 < y_3$, it is impossible that $y_2 \leq x_1 + x_2 - y_1$ since $y_2 \geq y_3$. Thus, $y_2 \geq x_1 + x_2 - y_1$, and, by Condition (3) in Lemma 4, $o(\langle P, Q \rangle) \lesssim R$.

Case 4. $y_1 \leq x_2$, $y_2 \leq x_1$, and $y_2 \leq y_1$.

The subcases that $y_2 \geq x_3$, $y_3 \geq x_2$, and $x_3 \geq x_2$ and that $y_2 \leq x_3$, $y_3 \geq x_2$, and $x_2 \leq y_2$ result in $x_2 = y_2$ which contradicts the assumption $x_2 > y_3$. Thus, we only need to consider the following two cases.

Case 4.1. $y_2 \geq x_3$, $y_3 \leq x_2$, and $x_3 \geq y_3$.

We can find that $y_2 \geq x_3$, $x_3 \geq y_3$, and $x_1 + x_2 - y_1 \geq y_3$ since $x_1 + x_2 - y_1 \geq x_1$ and $x_1 \geq y_2 \geq x_3 \geq y_3$ (see Figure 4(k)). By Condition (2) in Lemma 4, $o(\langle P, Q \rangle) \lesssim R$.

Case 4.2. $y_2 \leq x_3$, $y_3 \leq x_2$, and $y_2 \geq y_3$.

We can find that $x_1 + x_2 - y_1 \geq x_1 \geq y_2 \geq y_3$, $x_3 \geq y_2$ and $y_2 \geq y_3$ (see Figure 4(l)). Thus, Condition (4) in Lemma 4 is satisfied which implies $o(\langle P, Q \rangle) \lesssim R$. This concludes the proof of the lemma. $\square$

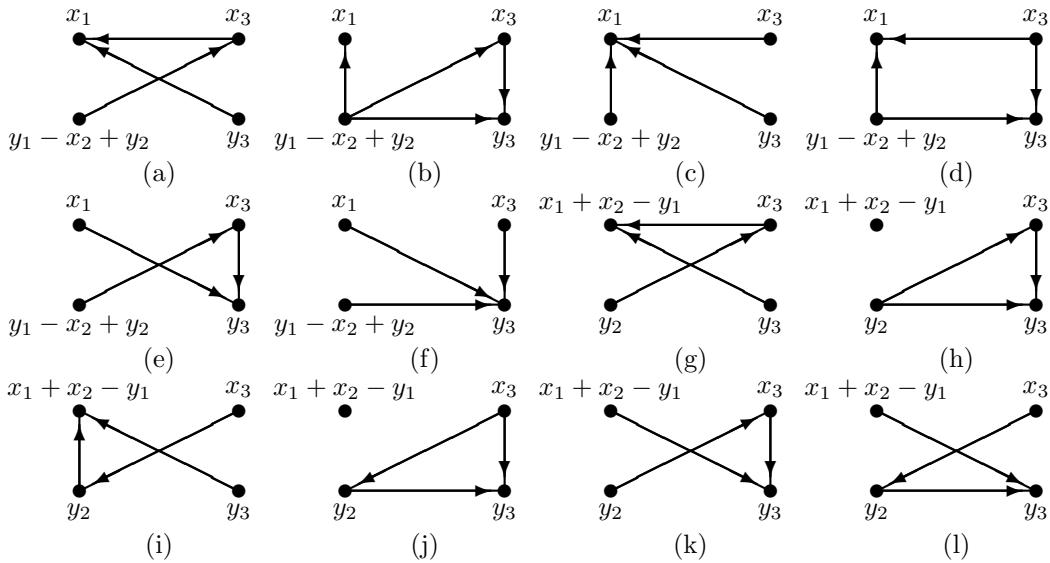


Figure 4: Illustrations for the cases in Lemma 12.

Lemma 13. If $P = (x, x)$, $Q = (x_2, y_2)$, and $P \preceq Q$, then $P \ominus Q = (x_2 + \Delta, y_2 + \Delta)$, where $\Delta = x - x_2$ when $x \geq x_2$ and $\Delta = 0$ otherwise.

Proof. By definition, if $x \geq x_2$, then $P \ominus Q = (x, x - x_2 + y_2)$; otherwise, $P \ominus Q = (x + x_2 - x, y_2) = (x_2, y_2)$. The former can be expressed as $P \ominus Q = (x, x - x_2 + y_2) = (x_2 + \Delta, y_2 + \Delta)$ with $\Delta = x - x_2$ and the latter can be expressed as $P \ominus Q = (x_2 + \Delta, y_2 + \Delta)$ with $\Delta = 0$. This completes the proof. $\square$

Lemma 14. If $P = (x, x)$, $Q = (x_2, y_2)$, and $Q \preceq P$, then $Q \ominus P = (x_2 + \Delta, y_2 + \Delta)$, where $\Delta = 0$ when $y_2 \geq x$ and $\Delta = x - y_2$ otherwise.

Proof. By using a similar argument as in Lemma 13, this lemma follows. $\square$

Lemma 15. Let $S = \langle P_1, P_2, \dots, P_n \rangle$ be a sequence of n points in the plane with $P_j = (x_j, y_j)$, for $1 \leq j \leq n$, and $x_i = y_i$, for some $1 \leq i \leq n$. Then, the x -coordinate value of $o(S')$ is smaller than or equal to that of $o(S)$, where $S' = \langle P_1, P_2, \dots, P_{i-1}, P_{i+1}, \dots, P_n \rangle$, i.e., removing P_i from S .

Proof. Assume that $x_i = y_i$, for some $1 \leq i \leq n$. Let $o(\langle P_1, P_2, \dots, P_{i-1} \rangle) = (a_1, b_1)$ and $o(\langle P_{i+1}, P_{i+2}, \dots, P_n \rangle) = (a_2, b_2)$. If $b_1 \geq x_i$, then, by Lemma 14, $o(\langle P_1, P_2, \dots, P_{i-1} \rangle) \ominus P_i = o(\langle P_1, P_2, \dots, P_{i-1} \rangle)$. By Lemma 2, this implies that $o(S')$ and $o(S)$ are the same o -minus point. For the case where $b_1 < x_i$, by Lemma 13 again, $o(\langle P_1, P_2, \dots, P_{i-1} \rangle) \ominus P_i = (a_1 + \Delta, b_1 + \Delta)$ with $\Delta = x_i - b_1 > 0$. Furthermore, $o(S) = (a_1 + \Delta, b_1 + \Delta) \ominus (a_2, b_2) = (a_1 + \Delta, b_1 + \Delta - a_2 + b_2)$ if $b_1 + \Delta \geq a_2$; otherwise, $o(S) = (a_1 + \Delta, b_1 + \Delta) \ominus (a_2, b_2) = (a_1 + a_2 - b_1, b_2)$. The o -minus point of S' can be computed as follows: $o(S') = (a_1, b_1) \ominus (a_2, b_2) = (a_1, b_1 - a_2 + b_2)$ if $b_1 \geq a_2$; otherwise, $o(S') = (a_1 + a_2 - b_1, b_2)$. By comparing the corresponding x -coordinate values of $o(S)$ and $o(S')$, the lemma follows. $\square$

Assume that $\mathcal{P} = \{P_1, P_2, \dots, P_n\}$ is a set of n points in the plane with $P_i = (x_i, y_i)$, for $1 \leq i \leq n$, and assume that there is more than one point with $x_i = y_i$. By Lemmas 14 and 15, if there exist a point (a, a) and a point (x_j, y_j) with $y_j > a$, then the convergence point of $\mathcal{P} \setminus \{(a, a)\}$ is the same as the convergence point of $\mathcal{P}$. Thus, we can assume that if $\mathcal{P}$ has a point with $x_i = y_i$, then $x_i > y_j$, for all other points P_j in $\mathcal{P}$. By this assumption, there can be at most one point with $x_i = y_i$. In the following, we assume that, for any set $\mathcal{P} = \{P_1, P_2, \dots, P_n\}$ of n points in the plane with $P_i = (x_i, y_i)$, for $1 \leq i \leq$

n , there is at most one point P_i in $\mathcal{P}$ with $x_i = y_i$, and, in this case, $x_i > y_j$, for all other points P_j in $\mathcal{P}$. Moreover, we also assume that S is a sequence of these n points with $Q_1 \preceq Q_2 \preceq \dots \preceq Q_n$, and we say that S is a *nondecreasing sequence* of $\mathcal{P}$.

Proposition 16. If $P = (x_1, y_1)$, $Q = (x_1 + \Delta, y_1 + \Delta)$, and $R = (x_2, y_2)$, then the x -coordinate value of $P \ominus R$ is smaller than or equal to that of $Q \ominus R$, where $\Delta \geq 0$.

Lemma 17. Let $\mathcal{P} = \{P_1, P_2, \dots, P_n\}$ be a set of points in the plane. Then, the convergence point of $\mathcal{P}$ is $o(S)$, where S is a nondecreasing sequence of the points in $\mathcal{P}$.

Proof. We prove this lemma by induction on n . By definition, the basis holds for $n = 2$. Now we prove this lemma for $n > 2$. Assume that the convergence point of $\mathcal{P}$ is $o(S')$, where $S' = \langle Q_1, Q_2, \dots, Q_n \rangle$ is not a nondecreasing sequence. We may assume that $Q_i = (x_i, y_i)$ is the point with the smallest index i such that $Q_{i-1} \succ Q_i$. Let $S_1 = \langle Q_1, Q_2, \dots, Q_i \rangle$ and $S_2 = \langle Q_{i+1}, \dots, Q_n \rangle$. Clearly, there exists a nondecreasing sequence of the points in $\{Q_1, Q_2, \dots, Q_i\}$. Assume that $S_3 = \langle Q_1, Q_2, \dots, Q_j, Q_i, Q_{j+1}, \dots, Q_{i-1} \rangle$ is a nondecreasing sequence. By the induction hypothesis, $o(S_3) = (a, b)$ is the convergence point of $\{Q_1, Q_2, \dots, Q_i\}$. By Lemma 3, $o(S_1) = (a + \Delta, b + \Delta)$ with $\Delta \geq 0$. Let $S'' = \langle Q_1, Q_2, \dots, Q_j, Q_i, Q_{j+1}, \dots, Q_{i-1}, Q_{i+1}, \dots, Q_n \rangle$. Thus, $o(S'') = o(S_3) \ominus o(S_2) = (a, b) \ominus o(S_2)$ and $o(S') = o(S_1) \ominus o(S_2) = (a + \Delta, b + \Delta) \ominus o(S_2)$. By Proposition 16, the x -coordinate value of $o(S'')$ is smaller than or equal to that of $o(S')$. Since $o(S')$ is the convergence point of $\mathcal{P}$, $o(S'') = o(S')$. We can repeat the above process on S'' until S'' is a nondecreasing sequence. This completes the proof of this lemma. $\square$

Theorem 18. Let $\mathcal{P} = \{P_1, P_2, \dots, P_n\}$ be a set of points in the plane. Then, the convergence point of $\mathcal{P}$ can be obtained in $O(n \log n)$ time.

Proof. By Lemma 17, the convergence point of $\mathcal{P}$ is $o(S)$, where S is a nondecreasing sequence of the points in $\mathcal{P}$. It is clear that S can be obtained in $O(n \log n)$ time by using the sorting technique.

4. Concluding remarks

As pointed out in the introduction section, we can apply the convergence point problem to the competition number problem when the competition number of each component is already given. However, it is worthy to investigate how to obtain the competition number of a component.

For example, consider a cycle of four points, denoted C_4 . Removing any edge from C_4 needs one isolated vertex and no isolated vertex is generated when the edge is removed. This step can be represented as $(1, 0)$. The remaining graph is a 3-path. Then, removing a leaf edge from the path will consume another isolated vertex, generate one isolated vertex, and can be represented as $(1, 1)$. Next step of removing any other leaf edge can also be represented as $(1, 1)$. The removing of the last edge yields the vector $(1, 2)$ since one isolated vertex is needed and two isolated vertex are generated. By using the o-minus operation, the above steps can be summarized by $(1, 0) \ominus (1, 1) \ominus (1, 1) \ominus (1, 2) = (2, 2)$. That is, the number of isolated vertices overall needed is 2, and the number of isolated vertices finally obtained is also 2. If the second removed edge is not a leaf edge, then it yields $(1, 0)$. Since the remaining graph is two 2-clique, any edge removing results in $(1, 2)$. Thus, the result can be computed by $(1, 0) \ominus (1, 0) \ominus (1, 2) \ominus (1, 2) = (3, 3)$. It turns out that different edge-removing sequences yield different results. In the C_4 example, the convergence point of $\{(1, 0), (1, 1), (1, 2)\}$ is $(2, 2)$. Thus, the competition number of C_4 is 2. Any connected graph can be decomposed into a clique cover. If we can find the o-minus point for the clique cover, we can solve the competition number problem.

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